Investigation: The Convex Lens Formula

Name: _____ Class: _____ Date: _____

Aim: To verify the mathematical model 1/u + 1/v = 1/f for a converging (convex) lens.

Before you begin

- Understand the meaning of focal length for a converging lens.
- Understand how a real image is formed by a converging lens.
- Know how image distance changes when object distance changes.
- Be familiar with the lens formula: 1/u + 1/v = 1/f.

Equipment

Converging lens of known focal length and lens holder; ray box; illuminated cross-threads (object); screen (e.g. white cardboard); retort stand and clamp; metre stick.

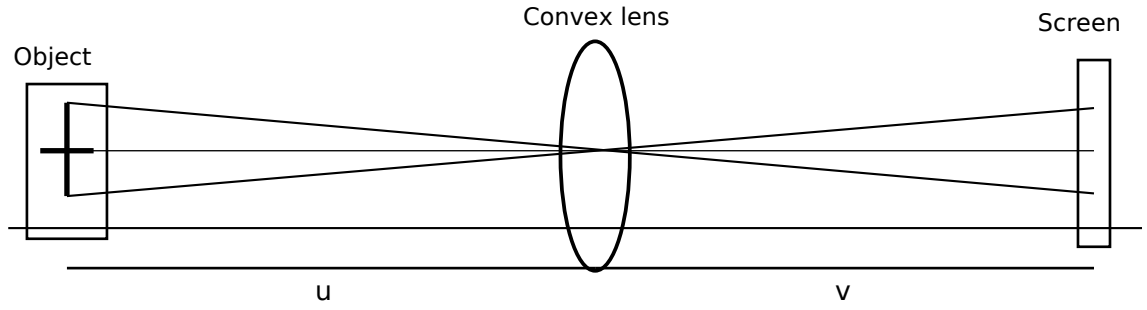
Safety

Wear safety glasses. Take care with the far end of the metre stick to avoid injury or damage.

Summary of method

Place the illuminated cross-threads in front of the lens. Move the screen on the opposite side until a sharp real image forms. Measure the object distance u and image distance v from the centre of the lens. Repeat for different object distances. Calculate f from each pair and compare the mean with the known focal length.

Apparatus arrangement



Experimental method

1. Estimate the focal length by focusing an image of a distant object (e.g. a window several metres away) onto the screen. The lens-to-screen distance is approximately the focal length.

2. Measure this distance with a metre stick and record it in the data table.

3. Set up the apparatus as shown. Place the illuminated object at a distance greater than the approximate focal length.

4. Adjust the screen until the image of the cross-threads is as sharp and clear as possible.

5. Measure u from the cross-threads to the centre of the lens and v from the real image on the screen to the centre of the lens. Record both measurements.

6. Change u and repeat steps 3–5 at least four more times. Include positions producing diminished and magnified images.

Results and numerical analysis

Approximate focal length from a distant object: _____ cm

Table of experimental data

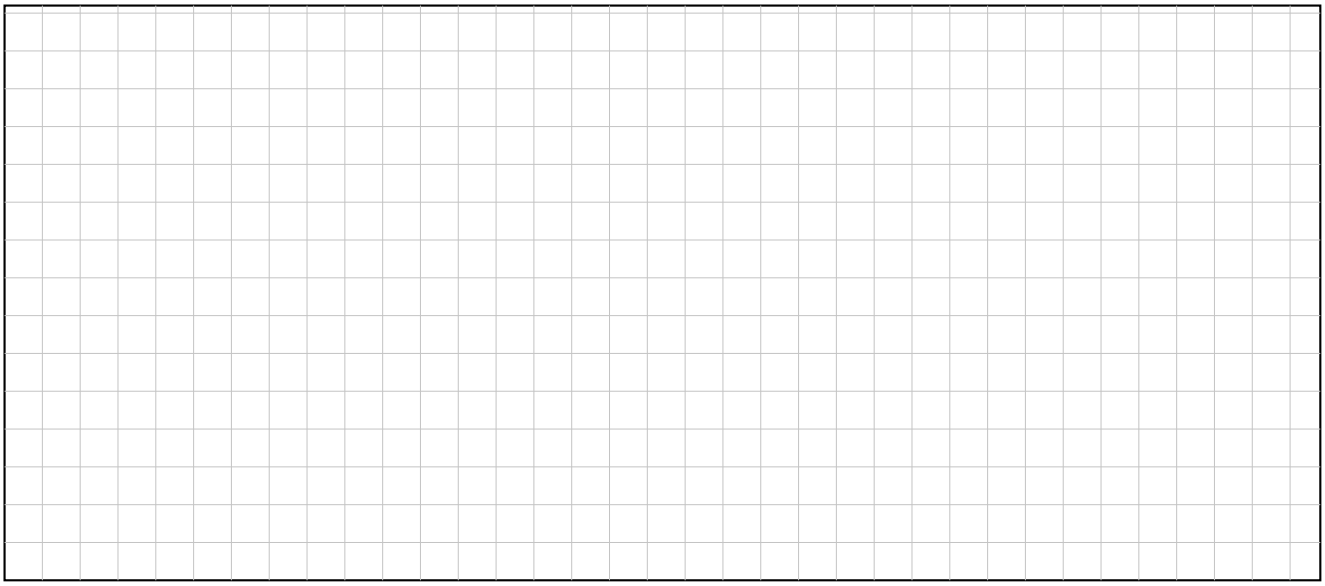
u (cm)	v (cm)	1/u (cm^{-1})	1/v (cm^{-1})	1/u + 1/v (cm^{-1})	f (cm)

Mean calculated focal length, f = _____ cm

Method 1: Processing the data numerically

1. For each pair of u and v, calculate 1/u, 1/v, their sum (1/f), and f.
2. Retain at least three significant figures for 1/u and 1/v.
3. Check whether the values of f are approximately the same.
4. Ignore clear outliers, calculate the mean f, and compare it with the known focal length.

Show your calculations (include units)

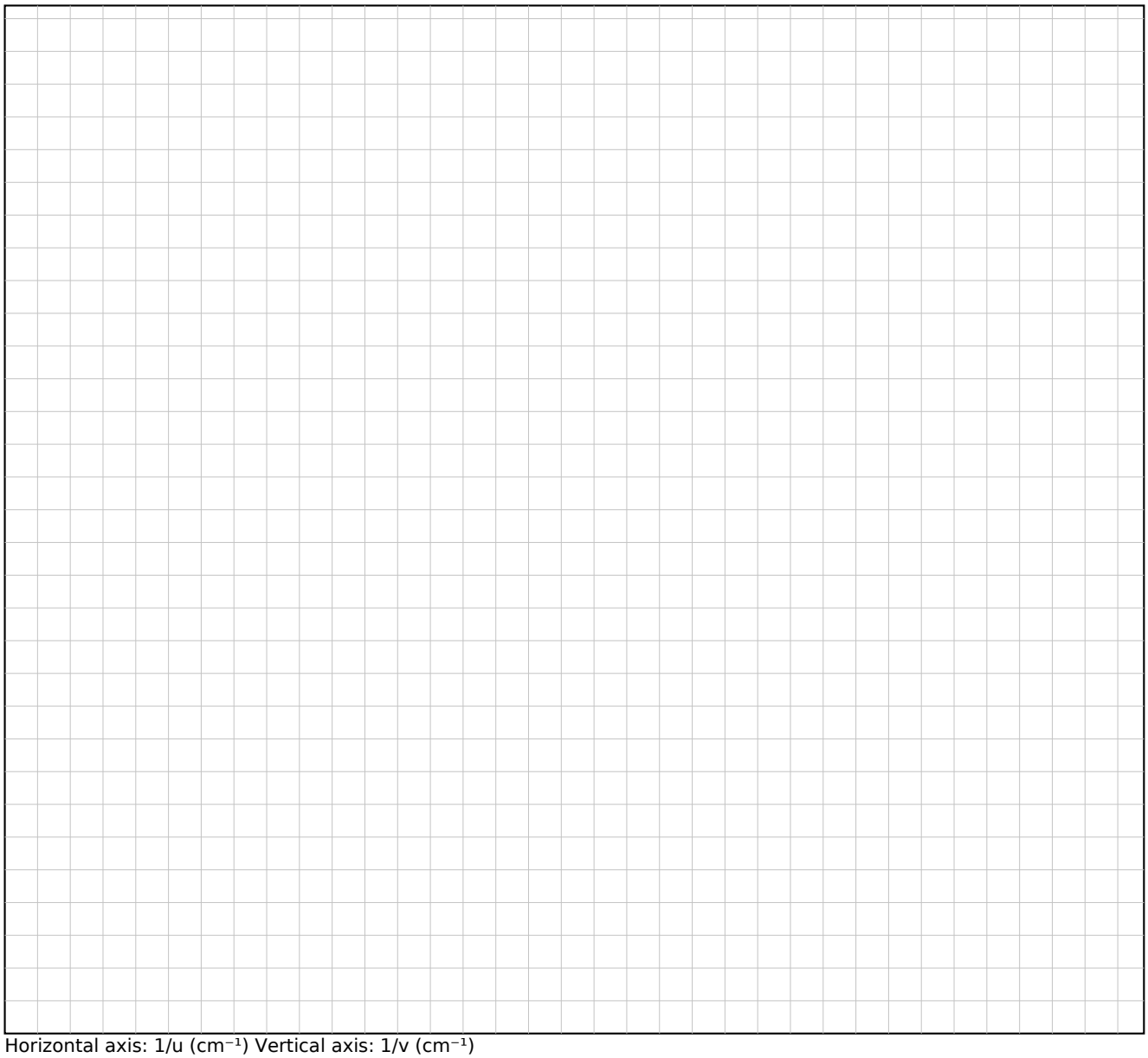


Graphical analysis

Method 2: Processing the data graphically

1. Calculate 1/u and 1/v for every measurement.
2. Plot 1/v on the vertical axis against 1/u on the horizontal axis.
3. Draw a straight line of best fit and extend it to meet both axes.
4. Read the values of 1/f at the vertical and horizontal intercepts.
5. Calculate the average intercept, then use it to determine f.
6. Compare this focal length with the numerical value and the known value.

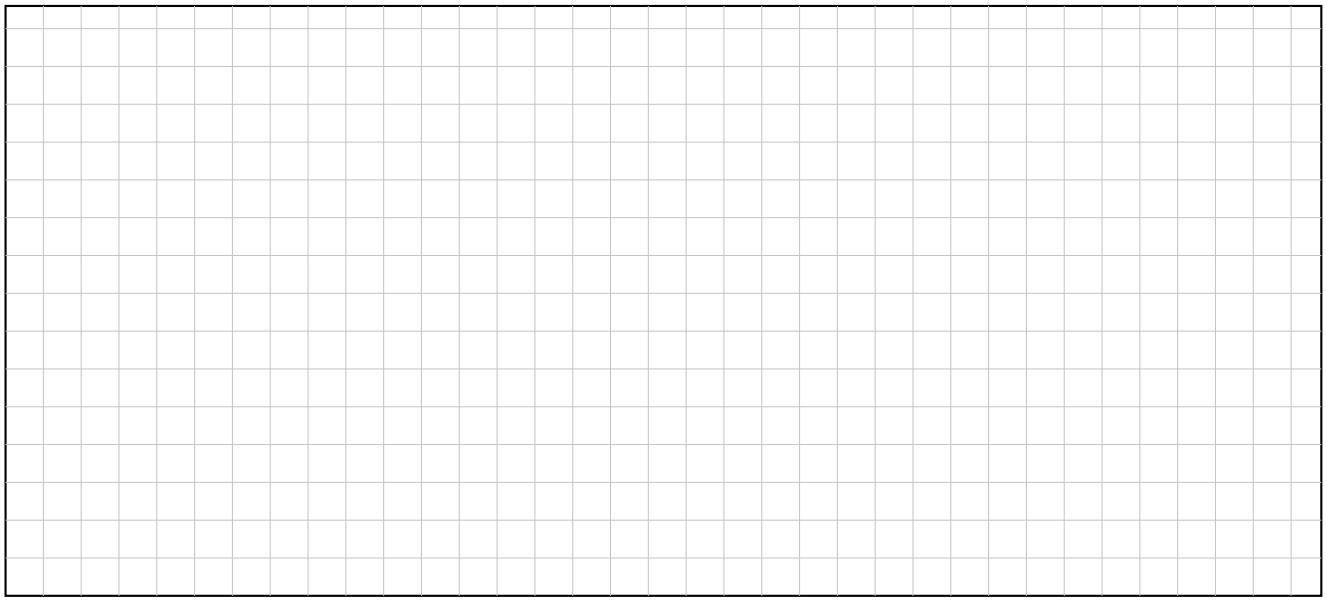
Graph: 1/v (cm^{-1}) against 1/u (cm^{-1})



Horizontal axis: 1/u (cm^{-1}) Vertical axis: 1/v (cm^{-1})

Graph calculations and conclusions

Calculations from the graph



Vertical intercept = _____ cm^{-1} Horizontal intercept = _____ cm^{-1}

Mean intercept = _____ cm^{-1} Graphical value of f = _____ cm

Explanation of the graph

The lens formula can be rearranged to $1/v = -1/u + 1/f$. A graph of $1/v$ against $1/u$ should therefore be a straight line with gradient -1 and intercept $1/f$ on each axis.

Conclusions from this experiment

Sources of error

- Deciding exactly when the image is in sharpest focus can introduce random error in v.
- Locating the optical centre of the lens is difficult and can affect measurements of u and v.
- Avoid parallax error when reading the metre stick.

Reflection questions

1. Is your calculated value of f close to the known (or estimated) focal length? Justify your answer.

2. Do your results verify that 1/u + 1/v = 1/f accurately represents image formation by a converging lens? Explain.

3. If you could not find the image on the screen, what might be wrong? How could you check?

4. Why might u be measured more accurately than v in this experiment?

5. List two precautions when measuring object distance u.

6. State one possible systematic error.

7. State the safety precautions you took.

8. Describe how you ensured accurate measurements.

9. Estimate the likely error in one of your measurements.

10. Were there any anomalous observations? How did you handle them?

11. How would you improve the accuracy and reliability of this experiment?

12. Would you expect similar results if you repeated the experiment? Explain.